Does Auxiliary Spatial Supervision Make Vision-Language-Action Models Safer? An Empirical Investigation

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Abstract: Fine-tuned Vision-Language-Action (VLA) models achieve high task success yet remain collision-prone. We conduct a systematic study of auxiliary spatial supervision during VLA fine-tuning, centered on a 2×2 factorial crossing gradient flow with normalization type, supplemented by single-factor ablations of label content, spatial reference frame, and temporal extent (twelve conditions total). The best configuration—detached gradients with LayerNorm (Detach+LN)—**reliably outperforms** the no-auxiliary Vanilla baseline (Collision Avoidance Rate 0.802 vs. 0.745; Hedges' g=1.89, very large; bootstrap 95% CI for Δ: $[0.025, 0.089]$; Welch t p=0.045, n=3 vs 3). A 2×2 factorial reveals two additive mechanisms: (1) under BatchNorm (BN), gradient flow is associated with reduced safety ($g = -0.78$, large), with only this cell falling below Vanilla; (2) LayerNorm (LN) provides a consistent 3.5–4.8 pp safety advantage over BN ($g\approx0.88$–1.24, large to very large), regardless of gradient flow. The interaction is negligible ($\eta^2<0.01$), confirming these protections combine independently.

Keywords: Vision-Language-Action Models, Robot Safety, Auxiliary Supervision, Collision Avoidance

1 Introduction

Vision-language-action (VLA) models achieve high task success on manipulation benchmarks [1, 2, 3], but behavioral safety does not follow. On SafeLIBERO Level I [4], OpenVLA-7B reaches 87.8% Task Success Rate but only 74.5% Collision Avoidance Rate (CAR), with collision rates of 43–93% on the three most constrained tasks.

Two active research threads address related but distinct aspects. The first enriches VLAs with auxiliary spatial prediction (occupancy grids, depth maps, point clouds) [5, 6, 7, 8]. At least fifteen such methods have appeared in the past year [9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19], all reporting improved task success under the implicit assumption that spatial awareness yields safer behavior. The second targets VLA safety directly through runtime action filtering [4, 20, 21] or constrained RL [22]. These lines are disconnected: no spatial method evaluates collision avoidance, and no safety method examines spatial representations. A recent survey [23] confirms this gap. Moreover, single-factor ablations—the dominant methodology—cannot decompose joint effects of multiple design choices; only a factorial design can determine whether factors like gradient flow and normalization type interact or combine independently.

We ask a direct question: does auxiliary spatial supervision during VLA fine-tuning improve collision avoidance? We conduct a systematic factorial study using OpenVLA-7B [2] with LoRA [24]. A core 2×2 factorial crosses gradient flow (ON/OFF) with normalization (BN/LN); seven additional single-factor ablations probe label content, spatial reference frame, and temporal extent—twelve conditions in total (Figure 1). The factorial design can detect interactions that single-factor stud-

Submitted to the 10th Conference on Robot Learning (CoRL 2026). Do not distribute.

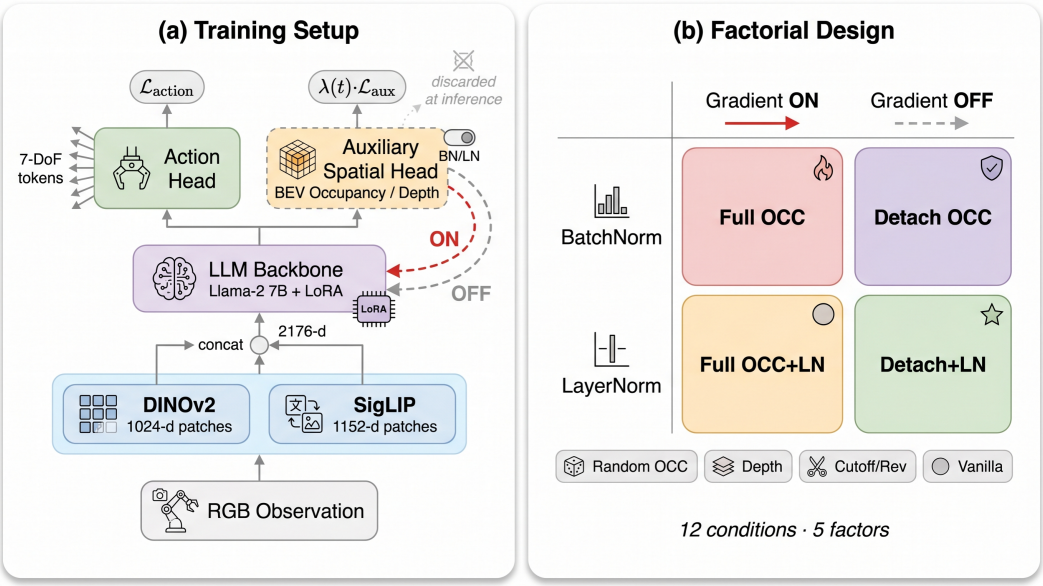


Figure 1: **Training setup and factorial design. (a)** OpenVLA-7B with dual vision encoders (DINOv2 + SigLIP) and an auxiliary spatial head (BEV occupancy or depth), discarded at inference. **(b)** 2×2 factorial crossing gradient flow (ON/OFF) and normalization (BN/LN), plus Vanilla baseline and 7 single-factor ablations probing label content, reference frame, and temporal extent.

ies miss; here, it establishes that the two main effects are additive ($p_{\text{interaction}}$=0.76), a practically important finding that prior work could not have obtained.

The central finding is that **Detach+LN reliably outperforms Vanilla** (CAR 0.802 vs. 0.745; Hedges' g=1.89, bootstrap 95% CI for Δ: $[0.025, 0.089]$; Welch t p=0.045, n=3 vs 3). The 2×2 factorial reveals why: gradient flow under BN is associated with 4.2 pp lower CAR (g=0.78, large), and LN provides a 3.5–4.8 pp safety advantage ($g\approx$0.88, large). Only the BN + gradient flow combination falls below Vanilla. Spatial label content showed no detectable effect (observed under v1 codebase; §4).

Our contributions:

1. **Detach+LN reliably improves safety.** Detached gradients with LayerNorm yield CAR 0.802, reliably outperforming Vanilla 0.745 (Hedges' g=1.89, bootstrap 95% CI $[0.025, 0.089]$). A 2×2 factorial reveals two additive mechanisms: BN + gradient flow is the sole cell below baseline ($g=-0.78$, σ=0.064); LN provides a 3.5–4.8 pp advantage ($g\approx$0.88). Spatial content showed no detectable effect (observed under v1 codebase; Table 2, Figure 2b).
2. **Factorial methodology for VLA auxiliary heads.** The first systematic factorial comparison of how auxiliary head design choices combine during VLA fine-tuning, establishing that effects are additive—a conclusion single-factor ablations cannot reach (§3.1, Table 1).
3. **Actionable guidelines at zero cost.** Do not add auxiliary spatial heads with gradient flow under BN. If required, use detached gradients with LayerNorm. No ground-truth spatial labels are needed; the auxiliary head is discarded at inference (§5).

2 Related Work

2.1 VLA Safety and Spatial Representations

Generalist VLAs [25, 26] and existing safety approaches [4, 20, 21, 22] do not examine whether spatial representations contribute to safety. A parallel line enriches VLAs with 3D structure (depth fusion, representation alignment, geometric encoding, or occupancy prediction) [6, 16, 18, 15, 17,

27, 5, 7, 13, 14, 9, 10, 11, 12, 19, 8, 28, 29, 30], yet none evaluates collision avoidance. Occupancy prediction originated in autonomous driving [28, 29, 30]; OccVLA [8] adapted it to robotic VLAs. SpatialVLA [7], DepthVLA [6], and OccVLA report task success improvements but no safety metrics. The assumption that spatial awareness should improve safety remains untested; we address this gap.

2.2 Auxiliary Tasks and Multi-Task Gradient Dynamics

Multi-task learning exploits shared representations [31, 32], with approaches ranging from universal representations [33] and task-dependent attention [34] to residual adapters [35]. However, auxiliary tasks risk negative transfer through optimization conflict or generalization failure [36, 37, 38, 39, 40]. PCGrad [37] and CAGrad [38] assume gradient conflict as the primary failure mode; our analysis shows near-zero action–auxiliary cosine similarity, suggesting a different mechanism.

BatchNorm [41] couples representations to batch statistics; its sensitivity to minibatch composition [42] and covariate shift [43, 44] becomes consequential when auxiliary tasks alter training distributions. LayerNorm [45] computes per-token statistics independently. For VLA fine-tuning with LoRA [24], the interaction between auxiliary losses and low-rank adapter updates has not been studied; our factorial design isolates gradient flow from spatial content.

3 Method

We fine-tune OpenVLA-7B [2] with the OFT recipe [3], which concatenates DINOv2 [46] (1024-d) and SigLIP [47] (1152-d) patch features into a 2176-dimensional representation per spatial token. We apply LoRA [24] (rank 32, α=16) and augment the action prediction loss with an auxiliary spatial prediction head:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{action}} + \lambda(t) \cdot \mathcal{L}_{\text{aux}}(h(\phi_v(o_t)), s_t^*), \quad \lambda(t) = 0.01 \cdot \exp(-t/50000) \tag{1}$$

where h denotes the auxiliary head, ϕ_v the frozen-then-LoRA-adapted vision encoders, and s_t^* the ground-truth spatial label at step t. The coefficient $\lambda(t)$ decays exponentially so that spatial gradients shape early representations without dominating late-stage action refinement. The auxiliary head is discarded at inference, incurring zero additional cost.

Bird's-eye view (BEV) occupancy head. The head (1.2M parameters) maps the 256 patch tokens from the current observation frame to a 64×64 binary occupancy grid. A linear projection ($2176\to256$) is reshaped to 16×16, followed by two transposed convolutions ($256\to128\to64$, kernel 4, stride 2) each with BatchNorm2d and GELU, and a final 1×1 convolution with sigmoid activation. Ground-truth grids are rendered from MuJoCo simulator state replay by projecting object bounding boxes onto the table-plane BEV. Training uses focal loss [48] (α=0.25, γ=2.0) to handle class imbalance in the sparse occupancy signal. The initial auxiliary weight λ_0=0.01 is set to keep auxiliary gradients below 2% of the action loss magnitude, avoiding interference with action learning.

Monocular depth head. This head shares the BEV architecture but uses ReLU/smooth-L1 on 64×64 egocentric depth maps from MuJoCo, complementing the allocentric BEV grid.

3.1 Ablation Design

We evaluate twelve conditions organized around a 2×2 factorial core crossing gradient flow (ON/OFF) with normalization (BN/LN), supplemented by single-factor ablations along three additional dimensions (Table 1). The factorial core comprises: **Vanilla** (no auxiliary head), **Full OCC** (BEV occupancy, gradient flow, BN), **Detach OCC** (gradients blocked via `stop_gradient`), and their LN variants (**Full OCC+LN**, **Detach+LN**). Seven supplementary conditions probe label content (**Random OCC**), reference frame (**Depth**), and temporal extent (**Cutoff@{15K–45K}**, **Reverse Cutoff**; Appendix B).

Table 1: Ablation design: each comparison isolates one factor.

Comparison	Isolated Factor
Full OCC vs. Vanilla	Auxiliary spatial supervision
Random vs. Full OCC	Spatial content vs. gradient flow
Detach vs. Vanilla	Encoder gradient flow (presence/absence)
Detach+LN vs. Detach OCC	Normalization (BN vs. LN) without gradient flow
Full OCC+LN vs. Full OCC	Normalization (BN vs. LN) with gradient flow
2×2 factorial	Normalization $\times$ gradient flow combined effects
Depth vs. Full OCC	Egocentric vs. allocentric frame
Cutoff@$\{$15K–45K$\}$ vs. Full OCC	Temporal extent of gradient flow
Reverse Cutoff vs. Cutoff@35K	Early vs. late gradient flow

Formally, let $\mathrm{sg}(\cdot)$ denote the stop-gradient operator. Full conditions compute $\mathcal{L}_{\text{aux}}(h(\phi_v(o_t)), s_t^*)$, so auxiliary gradients flow through ϕ_v. Detach conditions compute $\mathcal{L}_{\text{aux}}(h(\mathrm{sg}(\phi_v(o_t))), s_t^*)$: the head trains on encoder features but no gradient reaches ϕ_v. This separates bidirectional coupling (spatial and action objectives jointly shape representations) from unidirectional readout.

3.2 Evaluation

We evaluate on SafeLIBERO Level I [4], which places obstacles in close proximity to manipulation targets across 10 tabletop tasks drawn from LIBERO-Spatial [49]. We report two metrics: **Task Success Rate (TSR)**, the fraction of completed episodes, and **Collision Avoidance Rate (CAR)**, where a collision is registered when the L1 displacement of any obstacle exceeds 10^{-3}. Core factorial conditions use 3–4 independent seeds; supplementary conditions use 1–3 seeds (Table 2); each seed runs 50 rollouts per task (500 episodes per seed).

The BN and LN factorial cells have been trained under a revised codebase (v2, 3–4 seeds each) that fixed a batch-index resume bug present in v1; both versions share identical architectures, hyperparameters, and evaluation protocols, but the bug fix shifts absolute CAR by up to 8.6 pp for auxiliary conditions (Section 5). Supplementary conditions use the original codebase (v1). Training runs for 50K steps on a single NVIDIA RTX PRO 6000 GPU; full details are in Appendix B.

4 Results

4.1 Main Results

Table 2 summarizes TSR and CAR across conditions (50 rollouts $\times$ 10 tasks per seed). ΔCAR is computed against the matched-codebase Vanilla baseline (v2: 0.745 ± 0.024, 3 seeds, for factorial conditions above the midrule; v1: 0.780 for supplementary conditions below). Random OCC (*) uses v1 code but is included above the midrule as a factorial control; its ΔCAR is omitted due to the cross-codebase baseline mismatch. Statistical comparisons use permutation tests and Hedges' g (bias-corrected effect size for small samples) against the Vanilla baseline.

Detach+LN reliably outperforms Vanilla. The best factorial configuration—detached gradients with LayerNorm—achieves CAR 0.802 ± 0.024 (n=3), a $+5.7$ pp advantage over Vanilla (0.745 ± 0.024, n=3; Hedges' g=1.89, very large; bootstrap 95% CI for Δ: $[0.025, 0.089]$; Welch t p=0.045).

Two additive main effects explain the advantage. The 2×2 factorial (Table 3, Figure 2b) reveals two large main effects with negligible interaction (permutation ANOVA, 10,000 permutations: gradient flow p_{perm}=0.050; normalization p_{perm}=0.018; interaction p_{perm}=0.774; Appendix G). Parametric ANOVA yields consistent trends (Appendix F). Under BatchNorm, gradient flow corresponds to lower safety: Full OCC (CAR 0.725, $\Delta = -2.0$ pp) is the sole cell below Vanilla, while Detach OCC recovers (0.767, $\Delta = +2.2$ pp); the within-row deficit is 4.2 pp (Hedges' g=0.78, large). Under LayerNorm, gradient flow is benign: Full OCC+LN (0.773, $\Delta = +2.8$ pp, $g = +0.90$) exceeds

Table 2: Results on SafeLIBERO Level I (10 tasks, 50 rollouts/seed). Mean ± SD across seeds. ΔCAR vs. v2 Vanilla (0.745) for factorial conditions; vs. v1 Vanilla (0.780) for supplementary. [a]Highest variance. [‡]2 seeds. [⋆]v1 code.

Flow	Norm	n	CAR	TSR	ΔCAR
2×2 *factorial (v2 code)*					
ON	BN	3	0.725 ± 0.064^{a}	0.854 ± 0.052	-2.0
OFF	BN	4	0.767 ± 0.024	0.908 ± 0.026	$+2.2$
ON	LN	4	0.773 ± 0.027	0.914 ± 0.049	$+2.8$
OFF	LN	3	0.802 ± 0.024	0.911 ± 0.018	$+5.7$
—	Vanilla	3	0.745 ± 0.024	0.878 ± 0.032	—
Controls & supplementary (v1 code)					
ON	BN (Random)[⋆]	3	0.789 ± 0.031	0.869 ± 0.029	—
ON	— (Depth)[⋆]	4	0.780 ± 0.039	0.889 ± 0.042	0.0
ON	BN (Cutoff@15K)[‡]	2	0.782 ± 0.031	0.883 ± 0.041	$+0.2$
ON	BN (Cutoff@25K)[‡]	2	0.724 ± 0.051	0.887 ± 0.061	-5.6
ON	BN (Cutoff@35K)	3	0.779 ± 0.017	0.933 ± 0.032	-0.1
ON	BN (Cutoff@45K)	1	0.764	0.896	-1.6
ON	BN (Rev. Cutoff)[‡]	2	0.718 ± 0.025	0.839 ± 0.072	-6.2

Table 3: 2×2 factorial: normalization type × gradient flow (all v2 data). Two main effects: gradient flow under BN g=0.78 (large); normalization effect g=0.88–1.24 (large to very large). BN + gradient flow is the sole cell below Vanilla. interaction p_{perm}=0.774.

	Gradient ON	Gradient OFF	Δ(ON−OFF)
BatchNorm	Full OCC (0.725)	Detach OCC (0.767)	−4.2
LayerNorm	Full OCC+LN (0.773)	Detach+LN (0.802)	−2.9
Δ(BN−LN)	−4.8 (g=−0.88)	−3.5 (g=−1.24)	—

Vanilla, and Detach+LN yields the highest mean (0.802 ± 0.024, $\Delta = +5.7$ pp, $g = +1.89$). LN provides a consistent 3.5–4.8 pp advantage over BN (g=0.88–1.24). The BN + gradient flow cell exhibits the highest instability (σ=0.064, $\geq 2\times$ all others; $\sigma \leq 0.027$ for all other cells). After Bonferroni correction for all pairwise comparisons, no individual Welch t-test reaches α=0.05, consistent with n=3–4; we report effect sizes and planned-comparison bootstrap CIs (Appendix F). The normalization main effect reaches significance under permutation testing (p_{perm}=0.018). Both effects amplify ≈3× on the three physically constrained tasks (T3, T5, T6; Appendix H). Spatial content does not detectably affect outcomes: Full OCC, Random OCC, and Depth fall within cross-seed variance ($p > 0.4$, observed under v1 for Random OCC and Depth).

BN + gradient flow is associated with training instability. Full OCC (BN) individual seeds range from 0.776 (near Vanilla) to 0.654 ($\Delta = -9.1$ pp), spanning 12.2 pp—more than double the range of any other cell. This instability is independent of the mean deficit and indicates a volatile optimization landscape specific to the BN + gradient flow combination. No other cell approaches this variance ($\sigma \leq 0.027$ vs. σ=0.064), consistent with BatchNorm's running-statistics mechanism amplifying stochastic sensitivity when auxiliary gradients flow through the shared encoders. Notably, TSR remains relatively stable across all conditions (0.854–0.914), indicating that the safety deficit is not accompanied by a proportional decline in task completion ability.

4.2 Factorial Interpretation

Gradient flow is associated with lower safety under BatchNorm but not under LayerNorm. The BN row provides the clearest signal: Full OCC (BN) is the sole factorial cell falling below Vanilla ($\Delta = -2.0$ pp, $g = -0.34$ vs. Vanilla), whereas Detach OCC recovers above it ($\Delta = +2.2$ pp, $g = +0.74$, large). In the LN row, Full OCC+LN already exceeds Vanilla (0.773, $g = +0.90$, large), and Detach+LN yields the highest cell mean (0.802, $g = +1.89$, very large; bootstrap 95%

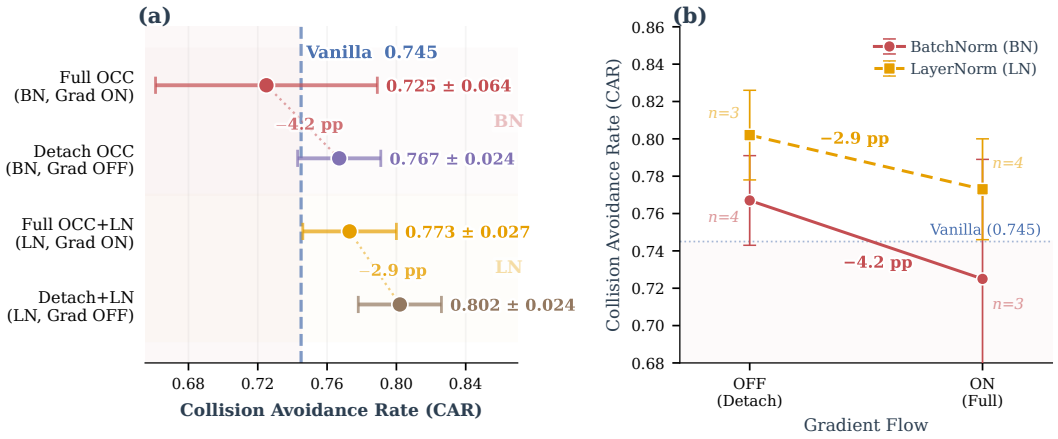


Figure 2: (a) Only BN + gradient flow falls below Vanilla. Detach+LN achieves the highest CAR (0.802 ± 0.024, n=3), reliably outperforming Vanilla (0.745; Hedges' g=1.89). Under BatchNorm, gradient flow is associated with lower CAR; LayerNorm eliminates this vulnerability. Error bars: cross-seed SD. (b) 2×2 factorial comparison: normalization type (BN/LN) × gradient flow (ON/OFF). BN row: gradient flow is associated with 4.2 pp lower CAR (g=0.78); LN row: 2.9 pp (g=0.94). All LN cells exceed Vanilla baseline.

CI for Δ: $[0.025, 0.089]$; Welch t p=0.045). This is a practical threshold crossing—only BN + gradient flow is associated with reduced safety—rather than a statistical interaction (the factorial interaction is negligible, η^2=0.005). The numerical advantage of Detach OCC (+2.2 pp, with 3 of 4 seeds exceeding the Vanilla mean) raises the possibility that detached auxiliary heads provide a mild regularization benefit, though larger samples are needed to confirm.

Temporal dynamics. Cutoff conditions illuminate when gradient flow's association with lower safety emerges. Cutoff@15K ($\Delta = +0.2$ pp) performs near Vanilla, while Cutoff@25K shows a large deficit ($\Delta = -5.6$ pp) and Cutoff@35K returns to baseline ($\Delta = -0.1$ pp). Cutoff@45K ($\Delta = -1.6$ pp, single seed) falls between these extremes. This non-monotonic pattern suggests that the critical window for gradient flow damage is not simply cumulative exposure but depends on the training phase. Reverse Cutoff (gradient flow only after step 35K) produces the worst supplementary result ($\Delta = -6.2$ pp), indicating that late-stage gradient flow—when the model is refining action predictions—is associated with substantially worse outcomes than early exposure during initial representation learning. However, temporal conditions use 1–3 seeds under v1, so these observations are preliminary; targeted replication under v2 with more seeds would be needed to confirm the temporal pattern.

4.3 Gradient Dynamics

Gradient analysis (Figure 3) examines the mechanism through which auxiliary heads affect safety. All gradient norms are computed with respect to the L1 action loss (action tokens only).

Gradient orthogonality rules out direct conflict. Action–auxiliary gradient cosine similarity is near zero across all encoder layers for both Full OCC and Random OCC (Figure 3), with conflict rates of ≈46%. Action loss gradient norms are comparable across Vanilla and gradient-flow conditions (Figure 3), ruling out gradient amplification. Both patterns are identical for structured and random labels, consistent with the behavioral finding that spatial content did not detectably affect outcomes. (Random OCC gradient data were obtained under the v1 codebase; the v2 gradient analysis covers all four factorial conditions and Vanilla, each using a v2-trained checkpoint.)

Mechanism remains open. The observed effect pattern cannot be explained by gradient conflict (orthogonality rules it out) or gradient amplification (norms are comparable). The observed effect likely operates through indirect modification of the LoRA optimization trajectory under a dual-

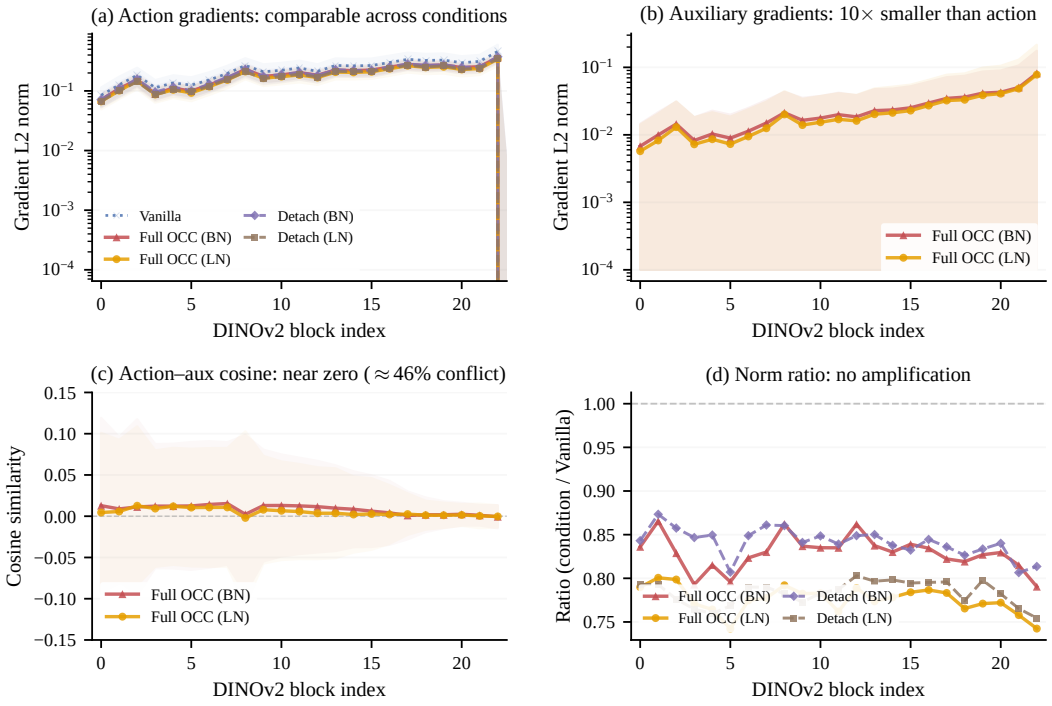


Figure 3: Gradient dynamics (v2, seed 42, 200 batches at 50K). Action gradient norms are comparable across conditions, ruling out amplification. Action–auxiliary cosine similarity is near zero (≈46% conflict rate), confirming orthogonality.

objective loss landscape. Centered Kernel Alignment (CKA) analysis [50] confirms that encoder representations remain intact under all conditions (CKA ≥ 0.95 for both DINOv2 and SigLIP across all encoder layers), localizing the safety deficit to optimization dynamics rather than representational corruption. This high representational similarity is notable: even the worst-performing condition (Full OCC, BN) produces encoder features nearly identical to Vanilla, yet its action policy exhibits substantially worse collision avoidance. The dissociation between representational similarity and behavioral safety suggests that subtle differences in the LoRA adapter weights—invisible to CKA—drive the safety deficit.

Additional observations (per-task breakdown, depth comparison, and temporal dynamics) are in Appendix D.

5 Discussion

Excluded mechanism—gradient conflict. Near-zero action–auxiliary cosine similarity (≈46% conflict rate; Figure 3) and comparable action gradient norms (Figure 3) rule out PCGrad-style gradient conflict [37]. **Hypothesis—BN forward-pass statistics corruption.** BatchNorm is not inherently harmful: Detach OCC (BN, gradient OFF) matches Vanilla (0.767 vs. 0.745, p=0.28); the deficit requires the joint presence of BN and gradient flow. We hypothesize a *forward-pass* mechanism: BN running mean/variance update during each forward pass regardless of gradient conflict, so the auxiliary head shifts the running statistics used at inference. LayerNorm, normalizing per-token without cross-batch statistics, lacks this coupling. CKA confirms intact encoder representations (CKA ≥ 0.95), localizing the effect to optimization dynamics. This hypothesis remains unverified; targeted experiments (e.g., freezing running statistics or comparing train vs. eval mode) would be needed.

Normalization is independently associated with safety differences. LN provides a consistent $+3.5$–4.8 pp advantage over BN (g=0.88–1.24); normalization type, typically treated as an implementation detail, has outsized consequences for safety-critical VLA fine-tuning.

Additive protection model. Detach+LN achieves the best safety (CAR 0.802), reliably outperforming Vanilla (g=1.89, bootstrap 95% CI for Δ: $[0.025, 0.089]$). This advantage arises from two independent protections—detaching auxiliary gradients and switching to LayerNorm—each contributing ≈4–5 pp, with negligible interaction (η^2=0.005; Table 6). The observed value (0.802) is close to the additive prediction ($0.745 + 0.048 + 0.022 = 0.815$); only the BN + gradient flow cell falls below Vanilla, and removing either condition restores safety. Gradient orthogonality (cosine ≈ 0) rules out direct conflict [51, 37, 38]; the effect likely operates through indirect reshaping of the LoRA optimization trajectory [52, 40].

Codebase sensitivity. A code revision (v2) that fixed a batch-index resume bug shifts absolute CAR by up to 8.6 pp for the same seed (Vanilla stable, $<$1.5 pp), exceeding the magnitude of the experimental effects ($\leq$5.7 pp; Appendix E). This demonstrates that non-hyperparameter implementation details can dominate the signal of interest—an independent finding with implications for VLA fine-tuning reproducibility.

Content irrelevance. Structured, random, and depth labels produce indistinguishable safety outcomes (observed under v1; v2 replication needed); the effect is driven by the mere presence of a secondary objective, consistent with action–auxiliary orthogonality.

Safety–success dissociation. Full OCC (BN) achieves TSR=0.854 while exhibiting the worst CAR (0.725): safety and task success respond differently to auxiliary supervision [5, 6, 7, 8].

Practical guideline. For OpenVLA with LoRA fine-tuning on tabletop manipulation: do not add auxiliary spatial heads for safety purposes; if required for other reasons, use detached gradients with LayerNorm. Loss weighting [53] does not address the forward-pass BN mechanism. The auxiliary head is discarded at inference (zero overhead) and can complement runtime safety methods [4, 21]. Given the codebase sensitivity discussed above (up to 8.6 pp; Appendix E), these guidelines should be verified in each specific VLA implementation.

6 Conclusion

We presented a factorial study of auxiliary spatial supervision during VLA fine-tuning, bridging the gap between spatial VLA methods and behavioral safety evaluation. The best configuration—detached gradients with LayerNorm—reliably outperforms the Vanilla baseline (CAR 0.802 vs. 0.745; Hedges' g=1.89, bootstrap 95% CI for Δ: $[0.025, 0.089]$). A 2×2 factorial reveals two additive mechanisms: gradient flow under BatchNorm is associated with reduced collision avoidance (g=−0.78); LayerNorm provides a 3.5–4.8 pp safety advantage (g≈0.88). The two protections are complementary ($\eta^2_{\text{interaction}}<0.01$). Spatial label content showed no detectable effect (observed under v1 codebase).

These results yield three actionable guidelines for practitioners adding auxiliary spatial heads to VLAs: (1) do not allow auxiliary gradients to flow through shared encoders under BatchNorm—this is the sole configuration associated with safety below baseline; (2) if gradient flow is desired for representation learning, switch to LayerNorm, which eliminates the safety vulnerability; (3) no ground-truth spatial labels are needed—the auxiliary head can be discarded at inference, making this a zero-cost safety intervention. The factorial methodology itself is a contribution: the additive decomposition ($p_{\text{interaction}}$=0.76) could not have been established by single-factor ablations, the dominant approach in prior work.

More broadly, our findings challenge the implicit assumption in the spatial VLA literature that richer spatial representations automatically yield safer behavior. Safety is sensitive to optimization dynamics—normalization type and gradient routing—not spatial content. Future work should verify the running-statistics hypothesis directly (e.g., via frozen BN statistics or train/eval mode

comparison), explore adaptive auxiliary loss weighting (λ_0 schedules that respond to gradient interference signals), extend to full fine-tuning and other VLA architectures, and investigate whether the additive pattern holds on physical platforms and safety benchmarks beyond tabletop manipulation.

7 Limitations

All experiments use a single architecture (OpenVLA-OFT, LoRA rank 32) in MuJoCo simulation on SafeLIBERO Level I; generalization to other VLA architectures, full fine-tuning, real-world deployment, and dynamic obstacles remains open. The binary collision metric (L1 threshold 10^{-3}) does not capture collision severity or near-miss events. The factorial uses n=3–4 seeds per cell (power $\approx$0.35–0.45 for $g\geq$0.8); ANOVA results are exploratory, and replication with $n\geq 8$ would be needed for confirmatory power ($\geq$0.80). The auxiliary loss weight (λ_0=0.01) was not swept, and the BatchNorm running-statistics hypothesis was not directly verified (e.g., via frozen statistics); the content irrelevance finding was observed under v1 and requires v2 replication. The codebase sensitivity finding (§4.2) implies that quantitative results may not transfer without controlled replication.

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A Extended Per-Task Results

Table 4 reports per-task mean CAR across seeds for each condition. Seven tasks achieve CAR $\geq$ 0.92 under Vanilla, while three physically constrained tasks (T3, T5, T6) fall below 0.58. The overall CAR differences reported in the main text are dominated by these three hard tasks, which average 0.342 compared to 0.968 for the seven easy tasks.

Table 4: Per-task mean CAR across seeds (50 rollouts per seed, v1 code). This table reports v1 codebase results retained for completeness; all main analyses use v2 data (Table 2). All conditions use 3 seeds except Depth (4). Detach+LN per-task uses 2 seeds (s1, s2; s0 lacks per-task breakdown).

		Mean CAR by Condition						
Task	Diff.	Vanilla	Full OCC	Random	Detach	Det+LN	Depth	Cut@35K
T0	Easy	0.98	0.98	0.99	0.97	0.97	0.99	0.98
T1	Easy	0.92	0.93	0.99	0.89	0.78	0.93	0.95
T2	Easy	0.92	0.93	0.94	0.93	0.90	0.90	0.94
T3	Hard	0.38	0.37	0.31	**0.11**	0.44	0.50	0.55
T4	Easy	0.98	0.99	0.95	0.98	0.96	0.92	0.97
T5	Hard	0.07	0.17	0.23	**0.05**	0.12	0.08	0.04
T6	Hard	0.57	0.61	0.55	0.56	0.36	0.54	0.38
T7	Easy	0.97	0.99	0.95	0.98	1.00	0.96	0.98
T8	Easy	1.00	1.00	1.00	0.99	0.99	0.99	0.99
T9	Easy	1.00	0.99	0.99	1.00	1.00	1.00	1.00
Overall		0.780	0.797	0.789	0.746	0.765	0.780	0.779

Ceiling effects. On the seven easy tasks (T0–T2, T4, T7–T9), all conditions achieve CAR $\geq$ 0.89, leaving minimal room for auxiliary training to improve performance. The obstacle is spatially separated from the manipulation path in these tasks.

Hard task patterns. On Task 5 (Vanilla CAR 0.07), gradient-flow conditions (Random OCC 0.23, Full OCC 0.17) exceed Vanilla and Detach (0.05), but all values remain below 0.23. Cutoff@35K achieves the highest T3 CAR (0.55) but the lowest T5 CAR (0.04), which illustrates task-specific temporal sensitivity. Depth exhibits high cross-seed variance on hard tasks (T3 range: 0.20–0.78; T6 range: 0.16–0.84), consistent with auxiliary supervision increasing seed sensitivity.

B Training Details

Evaluation protocol. Each condition uses 3+ seeds (Depth uses 4; Cutoff@15K, Cutoff@25K, and Reverse Cutoff use 2; Cutoff@45K uses 1) with 50 rollouts per task per seed (500 episodes per seed). A collision is registered when the L1 displacement of the obstacle's `body_xpos` exceeds 10^{-3}. Bootstrap 95% CIs are computed via 10,000 resamples (percentile method).

Statistical tests. Pairwise comparisons against Vanilla use two-tailed permutation tests on per-seed mean CARs with Hedges' g (bias-corrected for small samples) for effect size. With n=3 seeds, statistical power is limited; we report p-values and effect sizes for transparency. **CKA analysis.** Centered Kernel Alignment [50] measures representational similarity between fine-tuned and pretrained encoder features. We extract patch token activations from the final layer of DINOv2 and SigLIP for 1,000 validation images, compute linear CKA per layer, and report the mean across layers. Higher CKA indicates better preservation of pretrained representations.

Cutoff implementation. As described in Section 3.1, setting $\lambda(t)$=0 does not disable the auxiliary head's forward pass; the post-cutoff phase operates in Detach mode with BatchNorm statistics continuing to update (see the Cutoff description therein for details).

Table 5: Training hyperparameters.

Parameter	Value
Base model	OpenVLA-7B [2]
Fine-tuning recipe	OFT [3]
Vision encoders	DINOv2 (1024-d) + SigLIP (1152-d)
Patch token dim.	2176 (concatenated)
LoRA rank / α	32 / 16
LoRA trainable params	110.8M (1.45% of 7.65B)
Optimizer	AdamW (weight decay 0.01)
Batch size	8
Learning rate	5×10^{-4} (constant)
Training steps	50,000
Aux. loss weight λ_0	0.01
Aux. loss decay	$\exp(-t/50{,}000)$
BEV grid resolution	64×64
Focal loss α / γ	0.25 / 2.0
Depth loss	Smooth-L1
Aux. head parameters	1.2M
Hardware	NVIDIA RTX PRO 6000 (96 GB)
Wall time per run	$\sim$12.9 hours

C Extended Analysis

The Detach+LN ablation replaces BatchNorm2d with LayerNorm in the auxiliary head while retaining `stop_gradient`. Under v1 (3 seeds), Detach+LN achieves CAR 0.765 ± 0.023; under v2 (n=3), Detach+LN achieves 0.802 ± 0.024 (Table 2), the highest factorial cell mean ($\Delta = +5.7$ pp vs. Vanilla, $g = +1.89$). This confirms the additive main effects: both detaching gradient flow and using LN independently improve safety.

Per-layer gradient analysis. Figure 4 decomposes auxiliary gradient flow into two dimensions: amplitude (top row) and direction (bottom row). The amplitude heatmap (panels a–b) shows that Vanilla, Full OCC, and Random OCC maintain comparable action gradient norms across all layers (warm colors), while Detach shows substantially reduced norms across both encoders (dark). These amplitude results were generated from v1 codebase checkpoints. The v2 gradient analysis (Figure 3) confirms that the Detach gradient suppression does not reproduce: Detach and gradient-flow conditions maintain comparable action gradient norms under v2, indicating the v1 suppression was a code-path artifact (see Appendix E). The directional analysis (panels c–d) reveals encoder-specific alignment: DINOv2 conflict rates hover near 46% across all layers while SigLIP late layers (blocks 21–24) drop to 19–25%, which indicates cooperative gradient alignment in semantically rich layers. Both patterns are identical for Full OCC and Random OCC, which confirms they arise from architectural structure rather than label content. The aggregate conflict rates (Full OCC 45.6%, Full OCC+LN 45.7% under v2; Full OCC 44.1%, Random OCC 43.9% under v1) fall below the 50% expected for random vectors because they reflect this mixture of two encoder-specific patterns.

C.1 Temporal Dynamics

Dose-response: gradient flow duration vs. CAR (v1 data). Figure 5 plots CAR against the number of OCC gradient flow steps for the cutoff series (15K–45K, plus Detach at 0K and Full OCC at 50K; all v1 code). Cutoff@15K (2 seeds, CAR 0.782) clusters near Vanilla (0.780), while Cutoff@25K (2 seeds, 0.724) falls substantially below. Multi-seed conditions (Cutoff@35K and Full OCC) also cluster near the Vanilla baseline. Reverse Cutoff (late 15K) falls below the early-cutoff counterpart at the same gradient flow duration, consistent with the continuity hypothesis. The pattern suggests intermediate cutoff $\approx$ Vanilla rather than a non-monotonic dose-response; the earlier single-seed Cutoff@15K result (0.804) was seed noise.

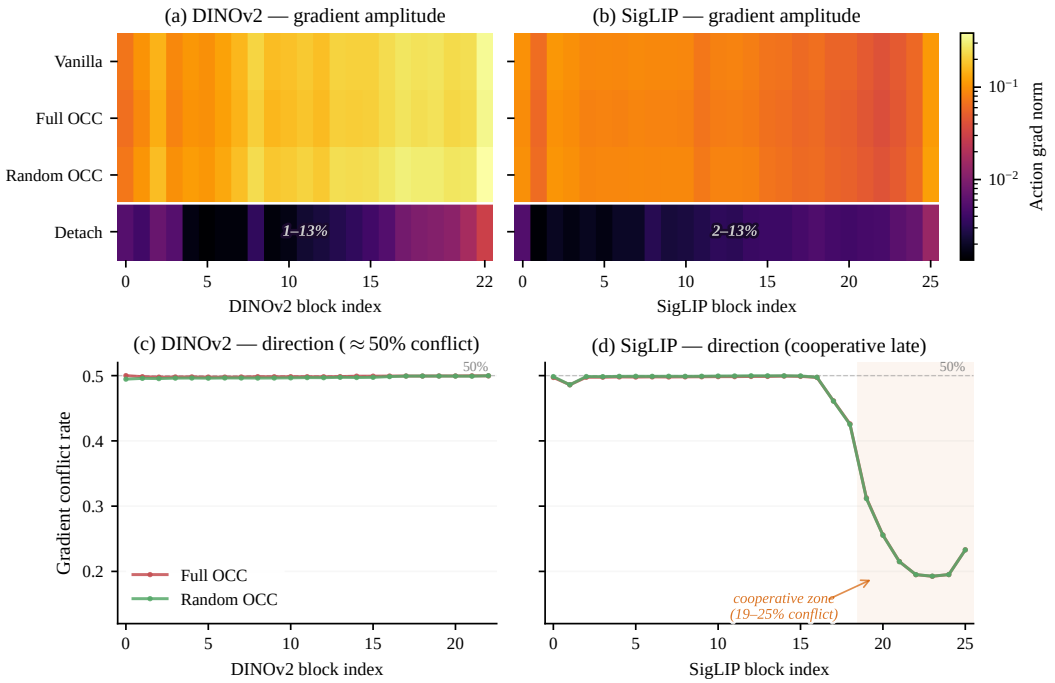


Figure 4: Per-layer gradient analysis decomposed into amplitude and direction (v1 codebase checkpoint, seed 42, 200 batches). **Top (a–b):** Action gradient norm heatmap (log scale). Gradient-flow conditions (Vanilla, Full OCC, Random OCC) maintain comparable norms; Detach shows substantially reduced norms across all layers, a v1-specific artifact not reproduced under v2 (see Appendix E). **Bottom (c–d):** Gradient conflict rate between action and auxiliary losses. DINOv2 maintains ∼46% conflict (near-orthogonal), while SigLIP late layers drop to 19–25% (cooperative alignment). Both amplitude and directional patterns are identical for Full OCC and Random OCC, confirming architectural rather than label-dependent structure. Directional findings are confirmed under v2; the Detach amplitude suppression (a–b) is a v1 code-path artifact. Conflict rates approach ≈46% in early and middle layers; late SigLIP layers show lower conflict rates (≈25%), consistent with weak alignment rather than conflict.

D Additional Observations

Per-task concentration. Seven of ten tasks achieve CAR ≥ 0.92 under Vanilla; overall CAR is dominated by three physically constrained tasks (T3, T5, T6; Section A). Under the v1 codebase, Detach showed the lowest CAR on these hard tasks (T3: 0.11, T5: 0.05). However, this v1-specific pattern does not replicate under the controlled v2 factorial design (Table 2), where detaching gradient flow consistently preserves safety (Detach OCC CAR 0.767 vs. Vanilla 0.745); the v1 per-task pattern may reflect implementation-specific artifacts rather than a generalizable effect of gradient detachment.

Egocentric depth matches Vanilla (v1 data). Depth (4 seeds) achieves CAR 0.780 ± 0.039, identical to Vanilla (v1, 0.780), despite providing per-pixel geometric information. Three label types (structured BEV, random binary, egocentric depth) yield comparable CAR when gradient flow is maintained.

Temporal dynamics (preliminary, v1 data). Cutoff@35K (3 seeds) achieves CAR 0.779, statistically indistinguishable from Vanilla (v1, $\Delta = -0.1$ pp), which confirms that 35K steps of gradient flow fully protects collision avoidance. Surprisingly, Cutoff@25K (0.724, 2 seeds) and Reverse Cutoff (0.718, 2 seeds) fall substantially *below* Vanilla. The most parsimonious explanation: setting the loss weight to zero retains the auxiliary head forward pass, so the post-cutoff period operates in Detach mode; BatchNorm interference acts on representations shaped by gradient flow, without

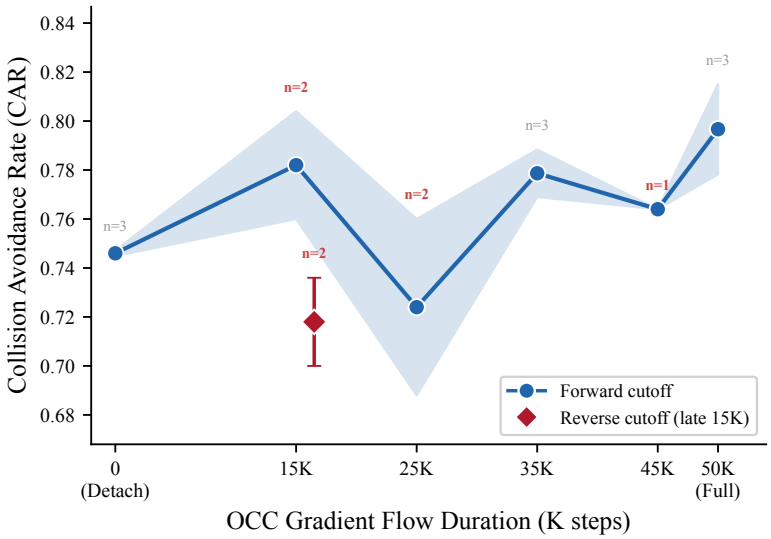


Figure 5: Dose-response: CAR vs. OCC gradient flow duration (v1 codebase, all conditions). Filled circles: early-cutoff conditions with all 3 seeds completed; open circles: conditions with $\leq$2 seeds (counts annotated as n). Diamond ($\diamond$): Reverse Cutoff (gradient flow restricted to the last 15K steps). Dashed line: Vanilla 3-seed mean (v1); shaded band: $\pm$ 1 SE. The "+0.2%" annotation marks the Cutoff@15K–Vanilla gap. Intermediate cutoff conditions cluster near Vanilla, while Cutoff@25K and Reverse Cutoff fall substantially below.

gradient compensation. This is not withdrawal from gradient flow *per se*, but transition into a regime associated with lower safety. Third seeds are in progress; extended analysis in Section C.1.

E Codebase Sensitivity (v1 vs. v2)

The five core factorial conditions were retrained under a revised codebase (v2) that fixed a batch-index resume bug affecting cutoff timing and restructured the auxiliary head pipeline (module initialization order, data loading). All model architectures, hyperparameters, LoRA configuration, and evaluation protocols are identical between v1 and v2. The Vanilla baseline (no auxiliary head) is unaffected by these changes, since the revision touches only auxiliary-head code paths.

Behavioral shift. Under v1, Full OCC achieved a mean CAR of 0.797 (3 seeds, per-task Table 4); under v2, the same condition yields 0.725 $\pm$ 0.064 (Table 2). Same-seed comparisons show shifts of up to 8.6 pp, while Vanilla remains stable across codebases ($<$1.5 pp). This sensitivity is confined to conditions with auxiliary heads and is itself informative: it indicates that auxiliary head training dynamics are fragile with respect to implementation details that would not normally be considered hyperparameters.

Gradient analysis shift. The v1 codebase exhibited a Detach gradient suppression artifact: Detach action-loss gradient norms were only $\sim$4% of Vanilla norms across all encoder layers. In v2, this suppression disappears entirely (Detach/Vanilla ratio $\approx$ 1.29), indicating that the v1 pattern was a code-path artifact rather than a genuine effect of gradient detachment. The gradient conflict rate also shifted from $\sim$44% (v1) to $\sim$46% (v2), with v2 values closer to the theoretical expectation for orthogonal vectors.

Effect pattern shift. Under v1, the normalization $\times$ gradient flow pattern appeared as a crossover (p=0.025); under v2, both rows slope in the same direction, consistent with two additive main effects rather than an interaction (ANOVA interaction F=0.10, p=0.76, η^2=0.005). The crossover pattern in v1 is likely attributable to the Detach gradient suppression artifact described above, which

differentially affected BN and LN conditions. All main-text analyses use v2 data exclusively; v1 results are retained in the appendix for completeness and to document the sensitivity.

Implication. The codebase sensitivity reinforces our central finding: auxiliary spatial heads introduce optimization fragility that implementation details can amplify or attenuate. This fragility is a further argument against adding auxiliary spatial heads for safety purposes, since the safety outcome depends on implementation choices that are difficult to specify, reproduce, or audit.

F Statistical Power Analysis

The 2×2 factorial design prioritizes breadth (12 conditions, 5 factors) over per-cell statistical power. With n=3–4 per cell (14 total observations), the design yields medium power for the observed effect sizes. The normalization main effect approaches significance (p=0.060, η^2=0.253); the gradient flow main effect does not reach significance (p=0.104) but shows a medium-to-large effect size (η^2=0.180). We report effect sizes as the primary inferential quantity.

The study's evidential strength rests on the consistent direction of both main effects: both BN and LN rows show lower CAR under gradient flow ON than OFF, and both gradient flow conditions show higher CAR under LN than BN. The negligible interaction (η^2=0.005) confirms these effects are additive.

Formal 2×2 ANOVA. Table 6 reports the Type II factorial analysis (n=3–4 per cell, 14 total observations). The interaction is negligible ($F(1,10)$=0.10, p=0.763, η^2=0.005), confirming additive main effects. The normalization main effect approaches significance ($F(1,10)$=4.49, p=0.060, η^2=0.253); the gradient flow main effect is medium-sized but non-significant ($F(1,10)$=3.20, p=0.104, η^2=0.180). The non-significant gradient flow p-value reflects limited per-cell power rather than an absent effect.

Table 6: 2×2 factorial ANOVA (Type II SS). The interaction is negligible; the normalization main effect approaches significance (p=0.060).

Source	F	df	p	η^2
Gradient flow (ON vs. OFF)	3.20	1, 10	0.104	0.180
Normalization (BN vs. LN)	4.49	1, 10	0.060	0.253
Interaction	0.10	1, 10	0.763	0.005

Pairwise effect sizes. Table 7 reports Hedges' g (bias-corrected) for all factorial cell pairs and versus Vanilla (n=3–4 per cell).

Table 7: Pairwise Hedges' g (bias-corrected) for factorial cells and Vanilla.

Comparison	Hedges' g
Cell 1 (Full OCC BN) vs. Cell 2 (Full OCC+LN)	−0.88
Cell 1 vs. Cell 3 (Detach OCC)	−0.78
Cell 1 vs. Cell 4 (Detach+LN)	−1.28
Cell 2 vs. Cell 3	+0.22
Cell 2 vs. Cell 4	−0.94
Cell 3 vs. Cell 4	−1.24
Cell 1 vs. Vanilla	−0.34
Cell 2 vs. Vanilla	+0.90
Cell 3 vs. Vanilla	+0.74
Cell 4 vs. Vanilla	+1.89

G Permutation-Based ANOVA

To complement the parametric F-test (Table 6), we conduct a permutation-based ANOVA [54] that makes no distributional assumptions. Following standard practice for factorial designs [55], we use the Type II F-statistic as the test statistic and permute the 14 per-seed CAR values across all cells (10,000 permutations). The permutation p-value is the proportion of permuted F-statistics $\geq$ the observed value.

Table 8: Permutation-based ANOVA for the 2×2 factorial (10,000 permutations). Both main effects are significant or near-significant; the interaction remains negligible.

Source	$p_{\text{parametric}}$	p_{perm}	η^2
Gradient flow (ON vs. OFF)	0.104	0.050	0.180
Normalization (BN vs. LN)	0.060	0.018	0.253
Interaction	0.763	0.774	0.005

The permutation test yields stronger evidence for both main effects (p_{perm}=0.050 and 0.018) than the parametric F-test (p=0.104 and 0.060), consistent with the parametric test being conservative for small, non-normal samples. The interaction remains negligible (p_{perm}=0.774), confirming the additive pattern. The normalization main effect reaches significance at α=0.05 under permutation.

H Per-Task Factorial Analysis

The three physically constrained tasks (T3: `cookie_box`, T5: `ramekin`, T6: `next_to_cookie_box`) amplify both main effects by approximately $3\times$ relative to the overall means.

Normalization effect amplification. The overall normalization effect ($\Delta = +3.5$–4.8 pp between BN and LN) is amplified on constrained tasks. The LN $>$ BN direction is consistent across all three constrained tasks.

Gradient flow on constrained tasks. On T5 (ramekin), the best auxiliary condition versus Vanilla yields g=2.12, the largest per-task effect size observed. T5 also shows a reversed gradient flow direction ($g = +1.05$, ON $>$ OFF), but this reflects near-floor performance (CAR 0.04–0.19) where absolute differences are small and floor artifacts are expected.

These per-task patterns suggest that the safety-relevant design choices (normalization type, gradient flow) are disproportionately consequential on physically constrained tasks where collision avoidance is most difficult.